

Multiple Choice (10 marks)

1. Which of the following rules regarding the interconversion of fuels are central to the integration of metabolic pathways?
 - (a) Fatty acids can be made from and converted to carbohydrates and amino acids
 - (b) Carbohydrates can be made from and converted to amino acids and fatty acids
 - (c) Fatty acids can be made from but not converted to carbohydrates and amino acids
 - (d) Amino acids can be made from and converted to fatty acids
2. Which nutrient intake is fairly similar for countries in north and south Europe?
 - (a) PUFA
 - (b) MUFA
 - (c) SFA
 - (d) Protein
3. Which of the following lipoproteins is produced via the exogenous lipoprotein pathway?
 - (a) High density lipoproteins
 - (b) Very low density lipoproteins
 - (c) Chylomicrons
 - (d) Low density lipoproteins
4. Fats normally contain triacylglycerols at concentrations of:
 - (a) 75-95%
 - (b) 50- 75%
 - (c) > 95%
 - (d) < 50%
5. In terms of price per calorie, which of the following foods is the cheapest and most nutritious?
 - (a) Apple
 - (b) Slice of bread
 - (c) Cheese burger
 - (d) Broccoli

True / False (10 marks)

Answer True or False

6. True or False: Insulin-like growth factor-1 is associated with an increased risk of prostate cancer.
7. True or False: Endurance type exercise training lowers liver glycogen content during prolonged activity.
8. True or False: Urea is the main nitrogenous compound found in human urine.
9. True or False: The worldwide prevalence of obesity is approximately 13%.
10. True or False: Weight loss greater than 10% or failure to regain birth weight by day 14 is concerning for a newborn's health.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the importance of screening for malnutrition upon hospital admission and analyze the effectiveness of the MUST tool in this process.
12. Discuss the role of vitamin B₁₂ in the body, particularly focusing on its function in the synthesis of methionine from homocysteine. Why is this process important for overall health?

End of Paper